

Q1 **DICTIONARY**

Trace the output:

```
d = {i: i*i for i in range(3)}  
print(sum(d.values()))
```

Your Answer

6

Q2 LIST

Compare `clear()`, `del`, and reassigning an empty list (`lst = []`). What are the differences in memory and references.

Your Answer

```
a=[10, 20, 30]
b=a
a.clear()
print(a)
print(b)
```

Q3 Tuple

Write a program to unpack a tuple into variables and also collect remaining values using *.

Your Answer

```
t = (10, 20, 30, 40, 50)
a, b, *rest = t
print(a, b)
print(rest)
```

Q4 LIST

2 marks

What are the key characteristics of Python lists? Explain dynamic resizing and heterogeneity with examples.

Your Answer

Python lists are ordered, mutable, and allow duplicate elements. They support dynamic resizing, meaning they can grow or shrink automatically as elements are added or removed. They are also heterogeneous, allowing a single list to contain different data types such as integers, strings, and floats simultaneously.

Q5 SET

2 marks

Explain properties of sets. Why are duplicates not allowed?

Your Answer

Sets are unordered collections of unique elements. Duplicates are not allowed because sets are designed to represent mathematical sets where each element must be distinct, and the underlying implementation uses a hash table to ensure each value is unique.